



# Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

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# The invisible layer

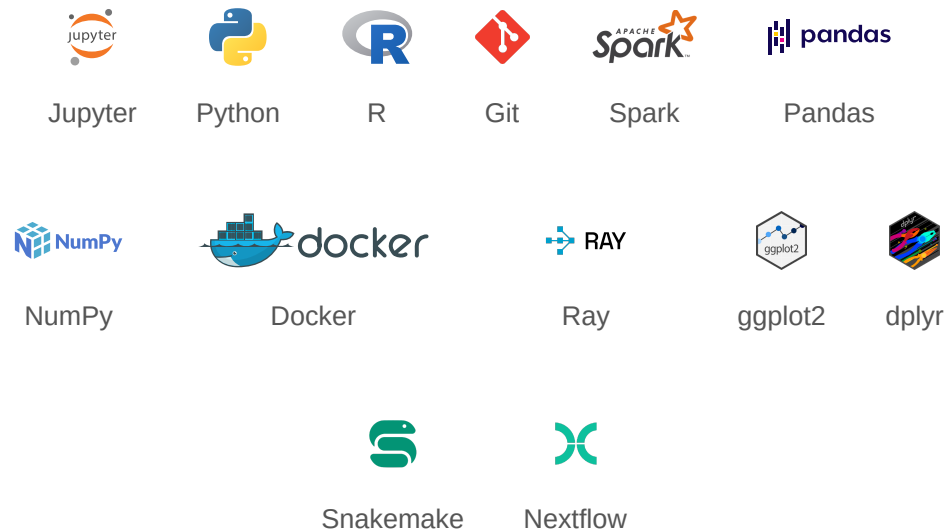
Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

**92%** of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.



# We've been here before

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People in this room helped build the open data movement:

- FAIR principles: from aspiration to institutional expectation
- Data management plans embedded in grant workflows
- Curation, provenance, and stewardship as professional services
- Open science infrastructure: repositories, PIDs, metadata standards
- **FAIR4RS**: FAIR principles now extended to research software

**Software is the next frontier**, and UC has been showing the way for 50 years: BSD, Unix, Ceph, Apache Spark, RISC-V, Ray. The pipeline from university research to global infrastructure is not rare or accidental.

“The research done in academia was the seed. Foundations provided the soil. But the community was the sun.”

## Summer 2000. UC Santa Cruz.

PhD student Jim Kent assembled the human genome on 50 borrowed PCs, icing his wrists to keep coding.

He finished **three days** before a corporate competitor's supercomputer.

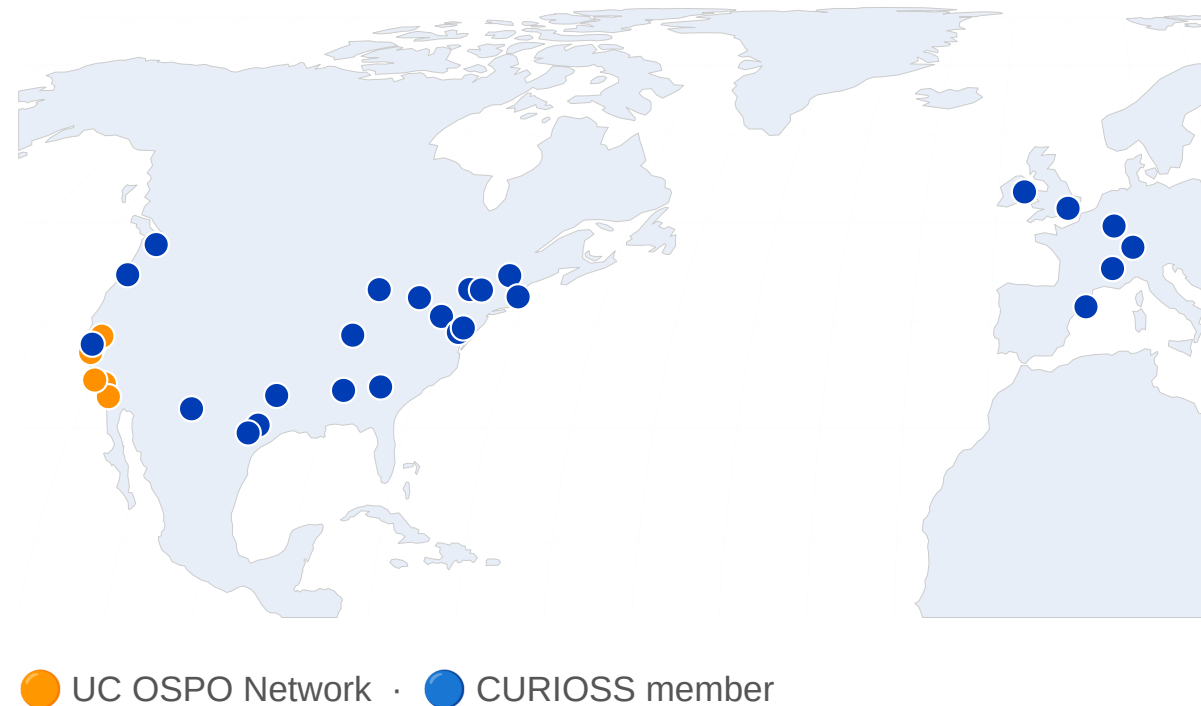
**The genome stayed in the public domain.**

Kent, J.W. (2002). BLAT - The BLAST-Like Alignment Tool. *Genome Research* 12, 656–664. · [genome.ucsc.edu](http://genome.ucsc.edu)

# What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPOs: Johns Hopkins, UC Santa Cruz, and RIT (2021)
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



# The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
- Launched April 2024, funded by the Alfred P. Sloan Foundation
- **\$1.85M** in external funding to date
- Lead: UC Santa Cruz, the first OSPO in a large state university system
- Serves 280,000+ students and 25,000+ faculty

Acts as a **neutral convener**: no single campus, company, or grant cycle can pull the work away.



# Three thematic areas

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## Discovery

Mapping who does what  
across the UC system:  
repos, contributors, tools,  
and practices



## Sustainability

Keeping projects and  
communities healthy  
through licensing, project  
health, and community  
support



## Education

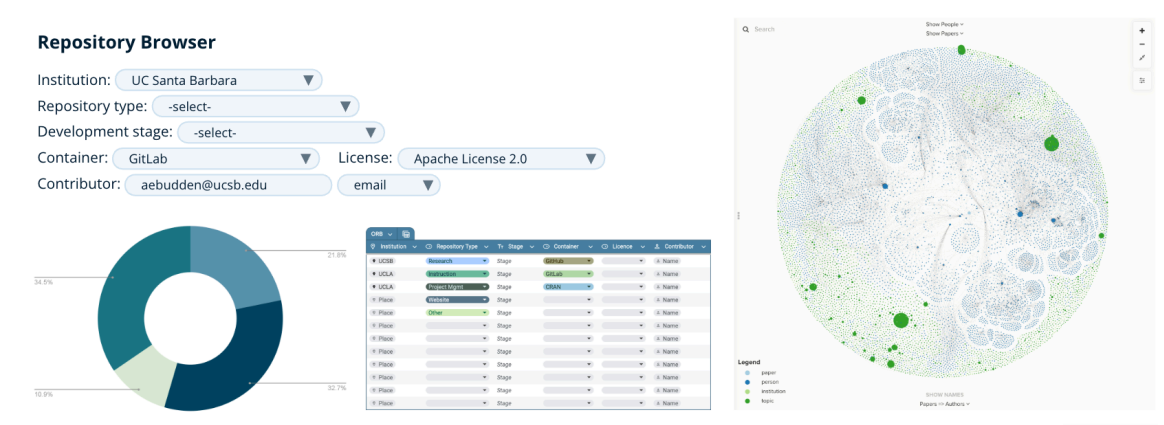
Coordinated training and  
learning pathways across  
campuses

# Discovery: knowing what's there

The 4 Ps framework, applied at system scale:

- **People:** who contributes to OS software at UC?
- **Product:** what has been built?
- **Practice:** how do faculty and staff engage?
- **Perception:** what are contributors' experiences?

Tools: GitHub pipeline (200,000+ repos scanned; ~52,000 institutionally affiliated across 10 campuses), network-wide survey (294 respondents), UC Open Repository Browser (UC ORB)



UC Open Repository Browser showing filter controls, repository type breakdown, data table of UC campus repos, and a network visualization of project relationships



# Sustainability: keeping it running

**58% of UC open source contributors are also maintainers.** Not users: stewards.

What they asked for most: sustainability grants, computing infrastructure, learning communities.

#1 challenge: **finding time to write documentation.**

Nationally: **60%** of maintainers are unpaid; **43%** report burnout.

Network services:

- Cross-campus licensing working group (UCOP + tech transfer + IT)
- Project health assessment (OSSPREY)
- Containerization support · Community management

## 58%

of UC OSS contributors  
are also **maintainers**  
not just users, stewards

## 60%

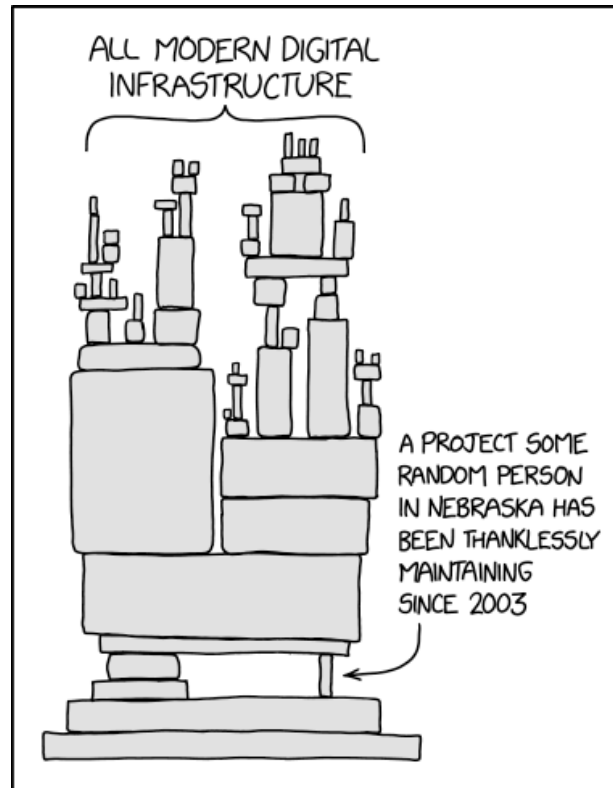
unpaid nationally

## 43%

report burnout

Tidelift (2024) State of the Open Source Maintainer Report

# The dependency problem



XKCD #2347 'Dependency': a towering stack of blocks labeled 'All modern digital infrastructure' balances on a single tiny block labeled 'A project some random person in Nebraska has been thanklessly maintaining since 2003'. By Randall Munroe.

xkcd #2347 (CC BY-NC 2.5) · UC survey: 58% of contributors are maintainers · Tidelift (2024): 60% unpaid, 43% burnt out

# AI and open source: a new pressure

AI code generation is changing what it means to maintain a project.

- **Slop PRs:** AI-generated contributions that look complete but are off-spec, shallow, or carry subtle bugs
- Review burden grows while maintainer capacity stays flat
- The OSS projects that trained the models are now flooded by their output
- Security risk: AI-generated code in dependencies with undetected vulnerabilities

**OSPOs are developing responses:** contribution disclosure policies, AI-aware code review norms, project-level guidance.

Baltes, Cheong & Treude (2026) arxiv:2603.27249 · GitClear (2024) “Coding on Copilot”

## The structural shift

**70%** of new AI PhDs go straight to industry

(a decade ago: 50/50)

**90%** of notable AI models come from a handful of companies

Academia is being sidelined from the systems it helped create.

Ruff (2026) UC Open Summit ·  
[youtu.be/eBriL3CDNeo](https://youtu.be/eBriL3CDNeo)

# Education: training as infrastructure

Gap analysis → curriculum inventory → learning pathways

- Inventoried existing materials: Carpentries, CodeRefinery, The Turing Way, UC-specific content
- Organized by topic (licensing, sustainability, community) and by role (contributor, maintainer, manager)
- Published at [ucospo.net/education](https://ucospo.net/education)
- Coordinated with data literacy programs and Carpentries infrastructure

UC OSPO NETWORK

- About
- Events
- Blog
- Resources
- Education
  - All Pathways
  - All Lessons
  - Browse Pathways
  - Search
  - Develop a Lesson
  - For Educators

## Welcome to the Open Source Learning Pathways.

What are you here to learn today?

**Getting Started with Open Source**  
For learners new to open source and open collaboration.

**Open Source Literacy**

**Beginner**  
GUIDE CONTRIBUTOR  
**Finding and Evaluating Projects**  
Learn how to find your first project using discovery tools, understanding labels, and identifying beginner-friendly tasks.  
Updated 12d ago

**Beginner**  
WORKSHOP COMMUNITY MANAGER, MAINTAINER  
**Social Coding and Inclusive Communication**  
Understand the 'human' side of open source, including respectful language, the 'release early, often' mindset, and the value of non-cod...  
Updated 3mo ago

**Beginner**  
WORKSHOP IP, GOVERNANCE  
**Software Licensing**  
This lesson provides an overview of common open-source software licenses, their conditions, and how to choose the right one for your project.  
Updated 7d ago

UC OSPO Network Education site showing the Open Source Learning Pathways, organized by role, skill level, and pathway, with beginner lesson cards including Software Licensing, Social Coding, and Finding and Evaluating Projects

# Distributed expertise, system-wide capacity

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Each campus brings what others don't. The network routes it to where it's needed.

## Each campus contributes

- Unique domain expertise and staff knowledge
- Local relationships with researchers and faculty
- Pilots that can only run with campus context
- Specialized capacity others can learn from

## The network multiplies it

- Expertise flows across all six campuses
- Shared staffing no single campus could fund
- Smaller campuses build capacity they couldn't alone
- A success at one becomes a template for five

Governed by an OSPO Leadership Group and thematic working groups, one per campus, coordinated at system level.

# Our seat is scholarly communication

We already promote best practices across the research lifecycle. Software is the same work in a new medium.

| What we do for scholarship | The software version            |
|----------------------------|---------------------------------|
| Citation and attribution   | software citation, CITATION.cff |
| Persistent identifiers     | Zenodo DOIs for releases        |
| Metadata and discovery     | repository metadata, FAIR4RS    |
| Licensing and rights       | open source license guidance    |
| Preservation               | versioned, archived releases    |

Not a new mission. The same one, pointed at code.

# Where this meets your work

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For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

## For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

# You're already doing this work

If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

**That is OSPO work.** The frame makes it legible to you, your institution, and the researchers you serve.

## Raise the floor, not just solve the ticket

Fixing their immediate problem gets them through the week.

Moving them from “it runs on my machine” to maintainable, licensed, citable software gets their whole lab there, and their students after them.

You don't need a formal OSPO to do this. You need the frame.

**OSPO heuristics already in your toolkit:** health signals · bus factor · governance files · FAIR4RS · software DMPs · contributor conventions · community governance



# Where it fits: instruction and outreach

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Two tracks, both already in our wheelhouse.

## Informational (the why)

- Why cite software, how to choose a license, planning for sustainability
- Workshops, consultations, guides, outreach

## Technical practice (the how)

- README, LICENSE, CITATION.cff, a versioned release, a DOI
- Hands-on: the UC OSPO citable-software lesson

We don't have to build an OSPO to take a seat. We teach the practices.

# What you can do right now

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If you advise on data-intensive research, you are probably already helping with code.

**The network has already built the resources** (Laura Langdon, UC OSPO):

- [README.md](#) - template + guide
- [LICENSE](#) - guide includes UC-approved options
- [CONTRIBUTING.md](#) - template + guide
- [CODE\\_OF\\_CONDUCT.md](#) - Contributor Covenant + guidance
- [CITATION.cff](#) + Zenodo release → citable, versioned DOI

## Going further:

- [Growing a Community Around a Project](#) - UC OSPO
- [Sharing Research Software: Reproducible, Citable, and Discoverable](#) - Carpentries Incubator lesson (covers [pixi](#), [CITATION.cff](#), Zenodo, discoverability)

# Learn more / connect

UC OSPO Network: [ucospo.net](https://ucospo.net)

- Education site: [ucospo.net/education](https://ucospo.net/education)
- Mailing list, Slack, events calendar



UC OSPO Network logo

Global network: [curiOSS.org](https://curiOSS.org), 34 academic OSPOs worldwide and growing

[ucospo.net](https://ucospo.net)

**Tim Dennis** Director, Data Science Center, UCLA Library [tdennis@library.ucla.edu](mailto:tdennis@library.ucla.edu)

Slides: [link TBD]

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